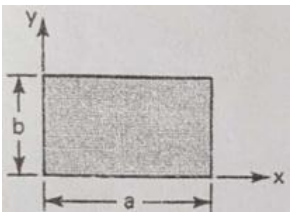
Higher National Diploma in Engineering (Civil/Mechanical),

2nd year, Second Semester Examination - 2022

ME 2217-Mechanics of Solids.

Question 01)

Consider a rectangular plate with sides a and b of thickness t.



a) Show that stress function $\boldsymbol{\phi}=-\frac{4}{b^3}XY^2\,(3b-2y)$ is a valid stress function. (06 marks)

b) Determine the stresses σ_x, σ_y and τ_{xy} for the above stress function. (06 marks)

c) Find the resultant normal and shearing boundary forces along all edges of the plate. (08 marks)

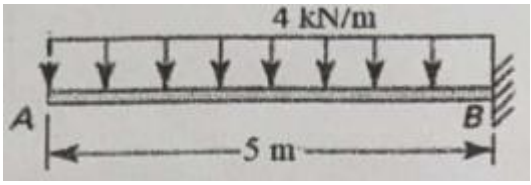
Question 02)

a) State Castigliano's Second theorem. (05 marks)

b) A cantilever beam AB is loaded with uniformly distributed load (4kN/m) throughout its length. Take E=100 GPa and I=1.5x10^{-6} m^4. (15 marks)

i. Determine the slope at the free end.

ii. Determine the deflection at the free end.



Question 03)

a) Derive Esler's equation to find the crippling bad of a culumn when both ends are fixed.(5 marks)
b) Write al ataumptions when yuyu derive the above equation. (3 marks)

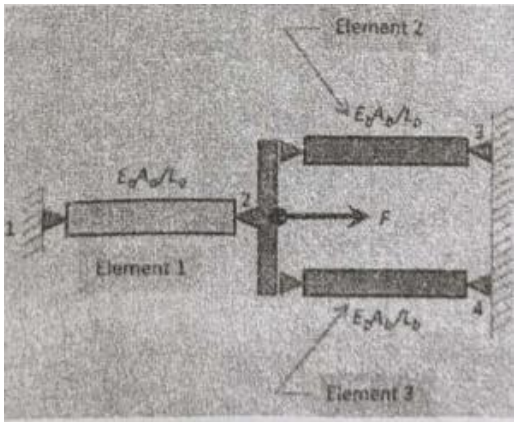
c) A hollow tube 4 m lang with.cotermal auf internal dameters of 50 mm and 25 mm respectively was found to extend 5 mm under a tensile lond of 65KN. Find,

i. the buckling load for the tube with both ends fixed (10 marks)

ii. the safe load on the tube taking a factor of safety as 3 (2 marks)

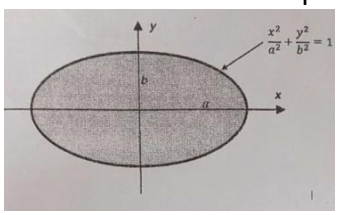
Question 04)

Determine the field values (nodal displacements and forces for the following structural system shown in figure. Take F-60 KN. A_a= 1.6 x 10^{-6} m² ,A_b=1.2 x 10^{-6} m^2 , E_a = 200 Gpa , E_b=70 Gpa , L_a =1m , L_a=1.5m



Question 05)

Figure shows the **elliptical cross-section** of a prismatic bar subjected to a **twisting moment** (T).



a) Obtain a suitable stress function. (04 marks)

b) Determine the angle of twist per unit length. (04 marks)

c) Derive the shear stress distribution of the bar. (04 marks)

d) Find the maximum value of shear stresses. (04 marks)

e) Derive an expression for warping function w(x,y). (04 marks)

Question 06

a) i) Draw the strain vs stress curve for a ductile test specimen under tensile test and label all important points of the curve. (03 marks)

ii) Briefly explain the process of structural failure into two pieces step by step using the above curve. (05 marks)

b) Write a short note on structural failure methods Fatigue and explain S-N cure used for fatigue test. (06 marks)

c)Write a short note on structural failure methods Creep and Fatigue. (06 marks)

d) Write a short note on structural failure method Buckling. (06 marks)